# Learning to Prefer Reliably: Error-Augmented Emotion Preference Optimization with Calibrated Fusion

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## Abstract

Emotion preference learning is important for aligning multimodal large language models (MLLMs) with human judgments of open-ended emotion descriptions and for training reward models that capture human emotional preferences. Given a video and two candidate descriptions, emotion preference prediction aims to identify the description that better matches the emotional state expressed by the video and its multimodal evidence. However, conventional pairwise supervision is often sparse, typically providing only a single negative description for each positive description, and therefore offers limited coverage of the diverse ways in which an emotion description can be incorrect. In particular, models may be insufficiently exposed to semantically plausible but emotionally inconsistent descriptions involving subtle errors in emotion category, polarity, intensity, or related semantic attributes. Moreover, judgments from an individual MLLM can be affected by model-specific biases when interpreting fine-grained or ambiguous multimodal emotional cues. To address these limitations, we propose **Error-Augmented Preference Optimization (EAPO)**, a progressive framework for improving the reliability of MLLM-based emotion preference judgment at both the data and model levels. First, we construct an error-augmented dataset by generating multiple controlled and emotion-aware negative descriptions from each positive description. We then adapt multiple independent MLLM judges to this richer supervision and aggregate their preference margins using margin-calibrated soft fusion, which maps heterogeneous margins to a common scale before aggregation. Experiments on the MER2026-EmoPrefer Challenge dataset and our error-augmented dataset demonstrate that EAPO improves emotion preference prediction and enhances the robustness of MLLM judges against semantically plausible but emotionally inconsistent descriptions.

## CCS Concepts

• **Computing methodologies** → **Machine learning**; **Artificial intelligence**; • **Human-centered computing** → *Empirical studies in HCI*.

## Keywords

MER2026, Preference Learning, Multimodal Emotion Preference Optimization



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## 1 Introduction

Descriptive multimodal emotion recognition represents affective states in free-form language rather than with a fixed label taxonomy. Such descriptions can express temporal dynamics, intensity, uncertainty, and evidence distributed across facial behavior, vocal delivery, linguistic content, and scene context [7, 9]. This flexibility complicates evaluation: a single reference description cannot enumerate every defensible interpretation, and lexical similarity does not necessarily reflect emotional faithfulness.

Emotion preference learning provides a tractable alternative for evaluating open-ended emotion descriptions. Given a video and two candidate descriptions, annotators compare the candidates and select the one that better reflects the observed emotional state. These comparisons support both the evaluation of descriptive multimodal emotion recognition systems and the preference-based alignment of affective models [4, 9]. The MER2026 Challenge formalizes this setting in the MER-Prefer track, where the task is to predict whether the first or the second description is preferred [7].

However, this formulation has two important limitations. First, conventional pairwise preference supervision is often sparse and under-specified. Each preferred description is typically paired with only one naturally occurring rejected description. Although this rejected description identifies which candidate is less preferred, it does not explicitly indicate why it is unreliable or ensure coverage of the different error modes that may occur. Such errors may involve an emotion flip, an intensity mismatch, a contradiction with observable evidence, or an incompatibility among multimodal cues. Consequently, preference judges may be insufficiently exposed to semantically plausible but emotionally inconsistent descriptions. Second, relying on a single MLLM judge makes the final decision dependent on that model's particular biases and failure patterns, especially when interpreting fine-grained or ambiguous multimodal emotional cues. These observations motivate a framework that improves both the coverage of error-relevant preference data and the diversity of judgment sources.

In this paper, we propose **Error-Augmented Preference Optimization (EAPO)**, illustrated in Figure 1, to improve the reliability of MLLM-based emotion preference judgment at the data and model levels. EAPO retains the original human-rejected descriptions as naturally occurring negatives and augments them with four controlled error categories: *Emotion Flip*, *Intensity Mismatch*, *Evidence Contradiction*, and *Modality Conflict*. We independently adapt multiple MLLM judges to this richer supervision through supervised fine-tuning followed by standard direct preference optimization. The judges do not communicate or exchange intermediate reasoning; each produces an independent preference signal. Once multiple signals are available, we combine them through margin-calibrated soft fusion. Specifically, judge-specific signed preference margins are mapped to a common scale before aggregation, preserving the graded strength of the preference signals that would be discarded by hard voting. EAPO therefore combines naturally occurring and controlled error supervision with independent multi-judge assessment and calibrated decision-level aggregation. Our main contributions are as follows:

- We propose **Error-Augmented Preference Optimization (EAPO)**, a framework that improves MLLM-based emotion preference prediction by addressing sparse negative supervision and dependence.
- We construct an error-augmented preference dataset that retains the original human-rejected descriptions and adds four controlled error types: *Emotion Flip*, *Intensity Mismatch*, *Evidence Contradiction*, and *Modality Conflict*. The resulting data are used to adapt MLLM preference judges through LoRA-based supervised fine-tuning and direct preference optimization, with candidate-order swapping to reduce position bias.
- We introduce margin-calibrated soft fusion for multi-judge preference prediction. The method maps preference margins from multiple independently adapted MLLM judges to a common scale before aggregation, thereby retaining graded preference evidence rather than relying on hard voting or direct averaging of raw margins.
- Experiments on the MER2026-EmoPrefer Challenge dataset and our controlled error-augmented subsets demonstrate the effectiveness of EAPO. Our submission ranked **sixth** on the MER2026 leaderboard and outperformed the official baseline models on the reported evaluation metrics.

## 2 Related Work

### 2.1 Multimodal Emotion Preference

Conventional multimodal emotion recognition maps visual, acoustic, and linguistic cues to predefined emotion categories [13, 16–23]. Although suitable for classification, categorical labels provide only a coarse summary and do not fully express the supporting cues, temporal changes, intensity, or uncertainty in a video. Descriptive emotion recognition instead represents affect in free-form language, allowing evidence from different modalities to be described jointly [8, 9]. Recent instruction-tuned MLLMs further support multimodal reasoning over such evidence [3]. However, several descriptions may be reasonable for the same video, making fixed-reference evaluation less suitable for open-ended emotion descriptions.

Emotion preference learning addresses this problem through pairwise comparison. Given a video and two candidate descriptions, an MLLM judge selects the description that is more consistent with human emotion preferences. EmoPrefer provides human-annotated preference pairs and evaluates both preference prediction and consistency [9]. MER-Prefer follows this binary formulation [7]. Building on this setting, we augment preference pairs with controlled description errors to improve preference prediction while retaining the ability to identify fluent but unreliable descriptions.

### 2.2 Preference Optimization

Preference learning uses comparative feedback when desired behavior is easier to rank than to specify directly. Early work learns reward functions from human comparisons and then optimizes a policy against the learned reward [4]. Bradley–Terry models provide a probabilistic account of such comparisons through the relative advantage of two candidates [1]. Direct preference optimization (DPO) avoids fitting a separate reward model and instead increases the relative likelihood of preferred responses with respect to a reference policy [10].

A related line of work studies language models as automatic judges. Text-oriented benchmarks have shown that model-based evaluation is useful but can exhibit systematic evaluator biases [12], and JudgeBench examines judge reliability on difficult response pairs requiring knowledge and reasoning [11]. MLLM-as-a-Judge extends this setting to vision-language inputs and evaluates agreement between multimodal judges and human annotations [2]. EmoPrefer further considers emotion judgments grounded in video, audio, and linguistic evidence [9]. EAPO uses standard SFT and DPO to adapt independent MLLM judges; its methodological contribution lies in controlled error augmentation and calibrated aggregation rather than a new preference objective.

## 3 Method

### 3.1 Problem Formulation

A MER-Prefer sample is represented as $q = (x, d_1, d_2, y)$, where $x$ denotes the multimodal evidence associated with a video, including visual, acoustic, and linguistic information when available. The two candidate emotion descriptions are denoted by $d_1$ and $d_2$, and the human preference label satisfies $y \in \{a_1, a_2\}$. Specifically, $a_1$ indicates that $d_1$ is preferred, whereas $a_2$ indicates that $d_2$ is preferred.

Given the multimodal evidence $x$ and the candidate pair $(d_1, d_2)$, a preference judge predicts either $a_1$ or $a_2$. The task aims to identify the human-preferred description by comparing both candidates against the visual, acoustic, and linguistic evidence conveyed by the video.

### 3.2 Error-Augmented Negative Construction

Preference judges are commonly trained on naturally occurring preference pairs, which may provide limited coverage of the errors that can appear in candidate emotion descriptions. Consequently, their performance may be less stable when unreliable descriptions exhibit error patterns that are underrepresented during training. To increase the diversity of negative examples, we augment the original preference pairs with controlled description errors.

For each labeled pair, we denote the human-preferred description by $d^+$ and retain the original human-rejected description. Starting from $d^+$, we use the Qwen3-Omni-30B-A3B-Instruct [15] to construct four additional descriptions, each corresponding to a controlled error type. Together with Original Rejected, they form the five negative types summarized in Table 1. We use $d_k^-$ to denote the negative description of type $k$, where $k \in \{1, \ldots, 5\}$. The resulting preference pairs are represented as $(x, d^+, d_k^-)$. Details of pipeline and dataset examples are provided in the Appendix.

The generated descriptions are constrained to remain fluent and plausible, avoiding superficial corruptions that would make the preference decision trivial. The five negative types cover complementary conditions, including original rejection, emotion flip,

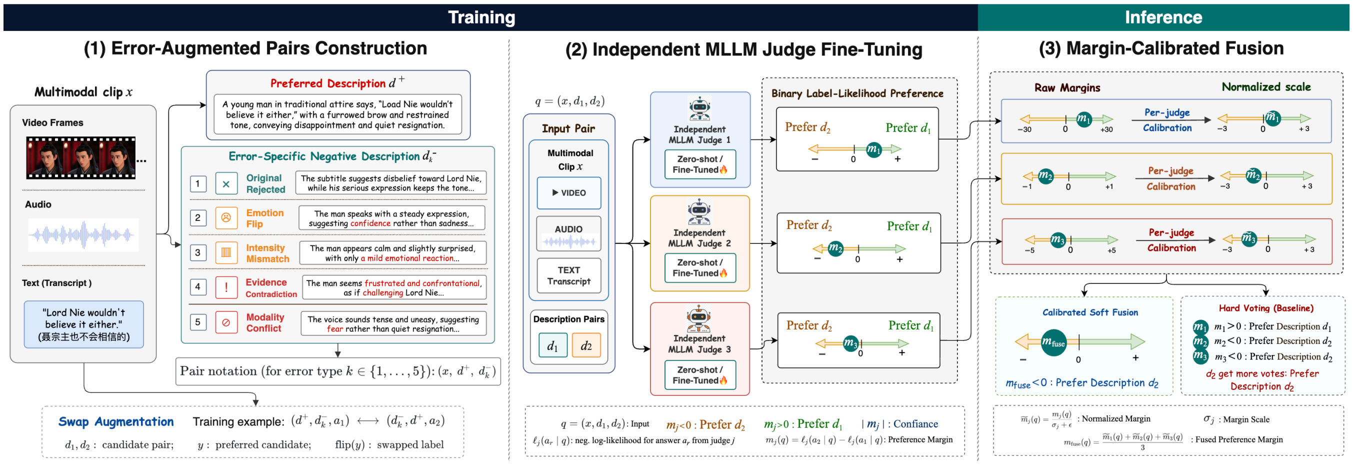


**Figure 1: Overview of our proposed EAPO framework. First, the error-augmented pair construction retains the original human-rejected description and introduces four controlled error types. Candidate-order swapping is used during LoRA-based SFT and subsequent DPO, with each MLLM trained independently. At the inference step, signed preference margins from different MLLMs are normalized to a common scale and combined through calibrated soft fusion to obtain the final preference.**

**Table 1: Five negative types used for error-augmented dataset construction.**

| No. | Error Type | Error Definition |
|---|---|---|
| 1 | Original Rejected | The naturally occurring non-preferred description in the original human-annotated preference pair. |
| 2 | Emotion Flip | Replaces the dominant affect with a different or opposing emotion. |
| 3 | Intensity Mismatch | Keeps the correct emotion label but overstates or understates its intensity. |
| 4 | Evidence Contradiction | Introduces a claim that is directly contradicted by an observable visual, acoustic, or linguistic cue. |
| 5 | Modality Conflict | Provides an interpretation that may fit one modality but ignores or incorrectly integrates conflicting evidence from other modalities. |

distorted intensity, evidence contradiction, and inconsistent integration of multimodal cues. Together, they expose the preference judges to a broader range of unreliable descriptions during training and support controlled analysis across different error types. The type labels are retained only as metadata for data organization and analysis; they are not provided to the preference judges.

## 3.3 Lightweight SFT-to-DPO Judge Adaptation

After constructing the error-augmented preference pairs, we adapt each MLLM judge independently through two successive stages: LoRA-based supervised fine-tuning (SFT) [6] followed by direct preference optimization (DPO) [10]. Both the original preference pairs and the four controlled error types are used during adaptation. For each pair $(x, d^+, d_k^-)$, the preferred and negative descriptions are placed in the two candidate positions, and the corresponding $a_1$ or $a_2$ label is used as the training target. We additionally swap the candidate order and update the target label accordingly to reduce position-dependent predictions.

During SFT, each judge is optimized with the standard language-modeling loss $\mathcal{L}_{\text{SFT}}$, which maximizes the likelihood of the correct answer label. This stage establishes the binary comparison format and adapts the model to compare emotion descriptions according to multimodal evidence. LoRA updates only a small set of trainable parameters, allowing multiple MLLM judges to be adapted independently with limited computational cost.

The DPO stage is initialized from the SFT checkpoint and optimized with the standard preference loss $\mathcal{L}_{\text{DPO}}$. For each training instance, the label corresponding to $d^+$ is treated as the preferred response, while the label corresponding to $d_k^-$ is treated as the rejected response. DPO increases the relative likelihood of the preferred label and further refines the preference boundary learned during SFT. Together, the two stages enable the judges to learn the emotion preference task while improving their discrimination between human-preferred descriptions and descriptions containing the constructed errors.

## 3.4 Margin-Calibrated Multi-Judge Fusion

We combine three independently adapted MLLM judges through Margin-Calibrated Multi-Judge Fusion. Given a comparison instance $q = (x, d_1, d_2)$, all judges receive the same multimodal evidence $x$ and candidate descriptions $(d_1, d_2)$. Here, $d_1$ and $d_2$ denote the two candidate descriptions, while $a_1$ and $a_2$ denote the corresponding answer labels. Specifically, $a_r$ indicates that the judge selects description $d_r$, where $r \in \{1, 2\}$.

For each instance, judge $j$ evaluates the two valid answer labels under the same prompt. We denote their negative log-likelihoods by $\ell_j(a_1 \mid q)$ and $\ell_j(a_2 \mid q)$, and define the signed preference margin as

$$m_j(q) = \ell_j(a_2 \mid q) - \ell_j(a_1 \mid q). \quad (1)$$

A positive margin indicates a preference for $d_1$, whereas a negative margin indicates a preference for $d_2$. The magnitude $|m_j(q)|$ reflects the strength of the judge-specific preference.

The numerical scales of raw margins may differ substantially across judges, making direct averaging unreliable. For a batch of $N$ comparison instances $\{q_n\}_{n=1}^{N}$, we estimate the standard deviation of each judge's raw margins as

$$\sigma_j = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(m_j(q_n) - \mu_j\right)^2}, \tag{2}$$

where $\mu_j$ is the mean raw margin of judge $j$ over the same batch. The statistics are computed independently for each judge using only its raw margins, without preference labels. We then apply the normalization:

$$\widetilde{m}_j(q) = \frac{m_j(q)}{\sigma_j + \epsilon}, \tag{3}$$

where $\epsilon = 10^{-6}$ ensures numerical stability. This transformation places the margin magnitudes of different judges on comparable scales while preserving their original preference directions.

Finally, the three normalized margins are combined with equal weights:

$$m_{\text{fuse}}(q) = \frac{1}{3}\sum_{j=1}^{3}\widetilde{m}_j(q). \tag{4}$$

The final preference prediction is determined by the sign of the fused margin:

$$\hat{y}(q) = \begin{cases} a_1, & m_{\text{fuse}}(q) > 0, \\ a_2, & m_{\text{fuse}}(q) \le 0. \end{cases} \tag{5}$$

Unlike hard voting, which retains only the discrete preference direction, the proposed soft fusion also incorporates the relative strength of each judge's preference after correcting for differences in margin scale.

# 4 Experiments

## 4.1 Datasets

Table 2 summarizes the datasets and their roles. We use EmoPrefer-Data-V2 for normal preference training and EmoPrefer-Data for normal validation [7]. After tie removal and binary filtering, these datasets contain 1,618 and 563 human-annotated preference pairs, respectively. The official evaluation comprises 379 samples in Stage 1 and 515 samples in Stage 2.

**Table 2: Dataset composition and experimental usage.**

| Dataset | Usage | Samples |
|---|---|---|
| EmoPrefer-Data-V2 | Normal Training | 1,618 |
| Error-Aug. Train-Set | Error-aug Training | 6,008 |
| EmoPrefer-Data | Normal Validation | 563 |
| Error-Aug. Val-Set | Error-aug Validation | 2,248 |
| MER-Prefer Stage-1 Test | Test Evaluation | 379 |
| MER-Prefer Stage-2 Test | Test Evaluation | 515 |

Error-Aug. Train-Set and Error-Aug. Val-Set contain only the four generated categories: *Emotion Flip*, *Intensity Mismatch*, *Evidence Contradiction*, and *Modality Conflict*. They exclude the naturally occurring Original Rejected pairs, which remain in EmoPrefer-Data-V2 and EmoPrefer-Data. After filtering, Error-Aug. Train-Set contains 6,008 generated pairs and Error-Aug. Val-Set contains 2,248 generated validation pairs. For error-augmented training, we combine EmoPrefer-Data-V2 with Error-Aug. Train-Set; for five-category validation, we combine EmoPrefer-Data with Error-Aug. Val-Set. Each combined set therefore covers one human-rejected category and four generated controlled-error categories. The generated sets are derived from the benchmark data and should not be interpreted as independently human-annotated benchmarks.

We report results on three evaluation sources. **Original Val** is the human-annotated EmoPrefer-Data validation split. **Error-Aug. Val-Set** is used during model development to evaluate performance on the four generated-error categories. Together, Original Val and Error-Aug. Val-Set cover the five negative-pair categories considered in this work. The **official test** consists of Stage 1 and Stage 2 and is reserved for final evaluation.

## 4.2 Experimental Setup

Following EmoPrefer [9], we consider two inference strategies. **Strategy 1 (S1)** directly provides the video and two candidate descriptions to the MLLM and asks it to select the description that better reflects the character's emotional state. **Strategy 2 (S2)** decomposes the judgment into two steps. The MLLM first generates a detailed multimodal description of the video; this generated description is then used as reference evidence when the model compares the two candidates. In our S2 implementation, the intermediate video descriptions are generated by Qwen3-Omni-30B-A3B-Instruct. Our adapted preference judges are based on MiniCPM-o-2.6-8B [5], Qwen2.5-Omni-7B [14], and Qwen3-Omni-30B-A3B-Instruct [15]. Table 3 additionally includes zero-shot results from GPT-5.5, GPT-5.5 Pro, MiMo-V2.5 under S1 and S2, and the Thinking and Instruct variants of Qwen3-Omni-30B-A3B under S2.

We use weighted F1 (WAF), the official ranking metric, to measure agreement with the preference labels. The **4-error Avg.** is the macro-average WAF over the four generated-error subsets. It measures the judge's average robustness to candidate descriptions containing the four controlled generated error types. **Swap Cons.** measures whether a model selects the same description identity after the two candidates exchange positions and is macro-averaged over the four generated-error subsets. Accuracy is an official secondary metric but is not reported in the following tables.

## 4.3 Overall Results

Table 3 shows the detailed results of both the original validation set and our generated error-augmented subset. First, performance on Original Val and the generated-error subsets does not always change in the same direction. This is most evident for Qwen3-Omni: training on the original preference pairs steadily improves Original Val, while both the 4-error Avg. and swap consistency decline. Preference accuracy on the original distribution is therefore insufficient to describe how reliably a judge handles plausible but incorrect descriptions.

Second, different judges provide complementary signals. Judge 16, based on Qwen2.5-Omni, reaches 79.75% on Original Val but is

**Table 3: Comparison of zero-shot, adapted, and fused preference judges on Original Val and the four generated-error subsets. Gray columns report aggregate results. Best and second-best results are shown in bold and underline, respectively.**

| ID | Model / Judges | Setting | Original Val | Emotion Flip | Intensity Mismatch | Evidence Contradiction | Modality Conflict | 4-error Avg. | Swap Cons. |
|---|---|---|---|---|---|---|---|---|---|
| 1 | GPT-5.5 | S1 Zero-shot | 66.34 | – | – | – | – | – | – |
| 2 | GPT-5.5 Pro | S1 Zero-shot | 66.56 | – | – | – | – | – | – |
| 3 | MiMo-V2.5 | S1 Zero-shot | 68.03 | – | – | – | – | – | – |
| 4 | MiMo-V2.5 | S2 Zero-shot | 67.18 | – | – | – | – | – | – |
| 5 | Qwen3-Omni-30B-A3B-Thinking | S2 Zero-shot | 72.37 | 88.73 | 87.09 | 93.31 | 92.96 | 90.52 | 84.21 |
| 6 | Qwen3-Omni-30B-A3B-Instruct | S2 Zero-shot | 73.18 | 92.08 | 90.92 | 96.17 | 96.09 | 93.82 | 91.55 |
| 7 | MiniCPM-o-2.6-8B | S1 Zero-shot | 59.48 | 86.92 | 82.03 | 92.70 | 92.17 | 88.46 | 81.94 |
| 8 | MiniCPM-o-2.6-8B | S1 Normal SFT | 61.77 | 87.01 | 81.93 | 92.53 | 92.53 | 88.50 | 81.67 |
| 9 | MiniCPM-o-2.6-8B | S1 Normal SFT+DPO | 61.43 | 86.83 | 81.94 | 92.44 | 92.44 | 88.41 | 81.81 |
| 10 | MiniCPM-o-2.6-8B | S1 Error-Aug SFT | 61.53 | 88.88 | 90.20 | 94.31 | 94.93 | 92.08 | 87.14 |
| 11 | MiniCPM-o-2.6-8B | S1 Error-Aug SFT+DPO | 63.27 | 89.29 | 90.38 | 95.10 | 94.84 | 92.40 | 87.46 |
| 12 | Qwen2.5-Omni-7B | S2 Zero-shot | 65.53 | 75.41 | 76.57 | 82.79 | 81.11 | 78.97 | 70.42 |
| 13 | Qwen2.5-Omni-7B | S2 Normal SFT | 75.84 | 86.45 | 89.21 | 93.06 | 92.61 | 90.33 | 84.91 |
| 14 | Qwen2.5-Omni-7B | S2 Normal SFT+DPO | 77.96 | 89.15 | 92.08 | 94.04 | 93.24 | 92.13 | 89.86 |
| 15 | Qwen2.5-Omni-7B | S2 Error-Aug SFT | 79.93 | 88.59 | 90.46 | 94.04 | 94.13 | 91.80 | 86.97 |
| 16 | Qwen2.5-Omni-7B | S2 Error-Aug SFT+DPO | 79.75 | 88.10 | 88.35 | 93.94 | 93.49 | 90.97 | 84.21 |
| 17 | Qwen3-Omni-30B-A3B-Instruct | S2 Zero-shot | 73.18 | 92.08 | 90.92 | 96.17 | 96.09 | 93.82 | 91.55 |
| 18 | Qwen3-Omni-30B-A3B-Instruct | S2 Normal SFT | 75.80 | 90.29 | 90.20 | 94.57 | 93.68 | 92.19 | 88.21 |
| 19 | Qwen3-Omni-30B-A3B-Instruct | S2 Normal SFT+DPO | 77.80 | 89.95 | 87.46 | 93.86 | 92.35 | 90.90 | 86.83 |
| 20 | Qwen3-Omni-30B-A3B-Instruct | S2 Error-Aug SFT | 75.46 | 87.22 | 87.56 | 93.23 | 92.34 | 90.09 | 83.90 |
| 21 | Qwen3-Omni-30B-A3B-Instruct | S2 Error-Aug SFT+DPO | 79.04 | **94.22** | **94.57** | **98.04** | **96.44** | **95.82** | **94.13** |
| F1 | Judges 16+19+21 | Calibrated Fusion | **80.82** | 92.25 | 93.23 | 96.71 | 95.20 | 94.35 | 91.41 |

weaker on the 4-error Avg. (90.97%) and swap consistency (84.21%). Judge 21 shows the opposite profile: its Original Val WAF is 79.04%, while its 4-error Avg. and swap consistency rise to 95.82% and 94.13%. Judge 19 adds a Qwen3 checkpoint trained only on the original pairs, providing a third decision pattern. Combining these judges raises Original Val WAF to 80.82%, above every constituent, while retaining a 94.35% 4-error Avg. and 91.41% swap consistency. The fusion result therefore reflects complementary model and training behavior rather than the strength of a single checkpoint.

MiniCPM provides further evidence that the effect is not limited to the two Qwen backbones. Error-Aug SFT+DPO improves its Original Val WAF from 59.48% to 63.27%, its 4-error Avg. from 88.46% to 92.40%, and its swap consistency from 81.94% to 87.46% relative to zero-shot evaluation. Although its absolute performance remains lower, the consistent improvement supports the use of error-augmented supervision across different judge architectures.

## 4.4 Effect of Error-Augmented Training

Table 4 provides a controlled comparison on Qwen3-Omni. Normal SFT raises Original Val WAF from 73.18% to 75.80%, and adding DPO further raises it to 77.80%. However, the 4-error Avg. falls from 93.82% to 92.19% and then to 90.90%; swap consistency follows the same downward trend, from 91.55% to 88.21% and 86.83%. Conventional adaptation therefore fits the original preference pairs more closely, but becomes less reliable when the rejected description contains a controlled semantic error.

The complete EAPO setting changes this pattern. Error-Aug SFT+DPO reaches 79.04% on Original Val, 95.82% on the 4-error Avg., and 94.13% swap consistency. Relative to Normal SFT+DPO, the improvements are 1.24, 4.92, and 7.30 points, respectively. Error-Aug SFT alone does not produce the same gains, showing that the generated pairs are most effective when used in the full SFT-to-DPO procedure. Within the scope of these constructed subsets, EAPO improves agreement with human preferences while making the judge more reliable against fluent descriptions that conflict with the multimodal evidence.

**Table 4: Impact of error-augmented training and SFT-to-DPO adaptation on Qwen3-Omni-30B-A3B-Instruct.**

| Setting | Orig. Val WAF ↑ | 4-error Avg. ↑ | Swap Cons. ↑ |
|---|---|---|---|
| Zero-shot | 73.18 | 93.82 | 91.55 |
| Normal SFT | 75.80 | 92.19 | 88.21 |
| Normal SFT+DPO | 77.80 | 90.90 | 86.83 |
| Error-Aug SFT | 75.46 | 90.09 | 83.90 |
| Error-Aug SFT+DPO (EAPO) | **79.04** | **95.82** | **94.13** |

The difference between the two training settings is also intuitive. The original pairs indicate which description was preferred, but provide limited coverage of the reasons why another description is unreliable. Normal SFT and DPO can therefore improve prediction on similar preference pairs without exposing the judge to a wider range of semantic mistakes. Error-augmented training supplies explicit examples of incorrect emotion, intensity, evidence, and cross-modal interpretation. The results are consistent with these additional contrasts helping the final DPO stage distinguish a well-supported description from an alternative that remains fluent but misrepresents the video.

## 4.5 Error-Type Robustness and Fusion Analysis

Table 5 compares selected checkpoints from the three model families with fusion over Judges 16, 19, and 21. Judge numbers refer to Table 3, and the error-set metrics use swap-balanced candidate orders to prevent candidate-position preferences from being mistaken for robustness to the generated errors.

Hard Voting first converts each judge's output into a binary decision and then follows the majority. This treats a weak preference

**Table 5: Comparison of single-judge selection, hard voting, and calibrated fusion on Original Val and the generated-error subsets. Best results are shown in bold.**

| Selected Judge(s) | Model / Strategy | Orig. Val WAF ↑ | 4-error Avg. ↑ | Swap Cons. ↑ |
|---|---|---|---|---|
| Judge 15 | Best Qwen2.5 judge | 79.93 | 91.80 | 86.97 |
| Judge 21 | Best Qwen3 judge | 79.04 | **95.82** | **94.13** |
| Judge 11 | Best MiniCPM judge | 63.27 | 92.40 | 87.46 |
| Judges 16+19+21 | Hard Voting | 79.22 | 94.50 | 91.59 |
| Judges 16+19+21 | Calibrated Fusion (Ours) | **80.82** | 94.35 | 91.41 |

**Table 6: Official Stage 1 and Stage 2 test results in WAF (%). Best results are shown in bold.**

| Type | Model / Judges | Setting | Stage 1 ↑ | Stage 2 ↑ | Macro ↑ |
|---|---|---|---|---|---|
| Official baseline | Qwen2-Audio | Zero-shot | 36.08 | – | – |
| Official baseline | Video-LLaVA | Zero-shot | 36.76 | – | – |
| Official baseline | LLaMA-VID | Zero-shot | 36.76 | – | – |
| Official baseline | LLaVA-Next-Video | Zero-shot | 41.31 | – | – |
| Official baseline | Qwen2.5-VL | Zero-shot | 76.77 | – | – |
| Official baseline | Qwen2.5-Omni | Zero-shot | 78.74 | – | – |
| Single judge | MiniCPM-o-2.6-8B | Judge 7 | 88.84 | 67.63 | 78.23 |
| Single judge | MiniCPM-o-2.6-8B | Judge 9 | 88.31 | 67.42 | 77.86 |
| Single judge | MiniCPM-o-2.6-8B | Judge 11 | 88.81 | 68.60 | 78.70 |
| Single judge | Qwen2.5-Omni-7B | Judge 12 | 84.94 | 68.47 | 76.70 |
| Single judge | Qwen2.5-Omni-7B | Judge 14 | 88.05 | 67.93 | 77.99 |
| Single judge | Qwen2.5-Omni-7B | Judge 16 | 87.79 | 69.11 | 78.45 |
| Single judge | Qwen3-Omni-30B-A3B | Judge 17 | 90.07 | 68.44 | 79.26 |
| Single judge | Qwen3-Omni-30B-A3B | Judge 19 | 90.43 | 66.93 | 78.68 |
| Single judge | Qwen3-Omni-30B-A3B | Judge 21 | 89.37 | 67.50 | 78.44 |
| Hard Voting | Judges 11+16+21 | Majority labels | 88.85 | 68.88 | 78.86 |
| Raw Fusion | Judges 11+16+21 | Raw-margin mean | 90.42 | 69.07 | 79.75 |
| Calibrated Fusion (Ours) | Judges 11+16+21 | Scale-normalized mean | **91.56** | **69.51** | **80.54** |

and a confident preference as identical votes. Calibrated Fusion instead retains each continuous preference margin, maps the margins from different judges to comparable scales, and averages them. A confident judgment can therefore contribute more than an uncertain one without allowing a model with naturally larger raw margins to dominate the result.

This difference is reflected on Original Val. Hard Voting obtains 79.22% WAF, while Calibrated Fusion reaches 80.82%, a gain of 1.60 points and the best result in Table 5. Their results on the generated-error subsets are very close: Hard Voting is higher by only 0.15 points on the 4-error Avg. and 0.18 points on swap consistency. Overall, the table shows that Calibrated Fusion provides the strongest validation preference result while maintaining almost the same error robustness as Hard Voting, supporting the use of calibrated continuous margins for the final decision.

## 4.6 Official Test Results

Table 6 reports WAF on the two official test stages; Macro WAF is their arithmetic mean. Official baselines provide Stage 1 results only. Judge numbers refer to Table 3; the fusion results combine Judges 11, 16, and 21. Qwen3 uses S2, while the available MiniCPM and Qwen2.5 predictions use S1.

The official Stage 1 baselines range from 36.08% to 78.74% WAF. All of our reported judges exceed the strongest baseline on this stage, although a direct Macro comparison is unavailable because the baselines do not report Stage 2. Across our individual judges, Qwen3 performs particularly well on Stage 1, with Judges 17 and 19 reaching 90.07% and 90.43%, respectively. Stage 2 is more difficult for every configuration: the individual results lie between 66.93% and 69.11%. This consistent gap affects the Macro scores and shows why performance should be reported on both stages rather than inferred from Stage 1 alone.

Judge 17 obtains the highest single-judge Macro WAF at 79.26%. Post-training does not improve every backbone in the same way. The zero-shot Qwen3 judge remains stronger than its adapted variants on the official test, whereas Error-Aug SFT+DPO gives MiniCPM its best family result and improves Qwen2.5 over its zero-shot setting. These differences suggest that the best adaptation recipe is model-dependent, even though error augmentation gives clearer gains on the controlled validation subsets.

The fusion results are more consistent. Hard Voting reaches 78.86% Macro WAF. Raw Fusion, which directly averages the unnormalized margins, improves this score to 79.75%, indicating that the margin magnitude contains useful information beyond the predicted label. Calibrated Fusion further increases Stage 1 to 91.56%, Stage 2 to 69.51%, and Macro WAF to 80.54%. It outperforms the best single judge by 1.28 points, Hard Voting by 1.68 points, and Raw Fusion by 0.79 points. The improvement over Raw Fusion supports the need to align judge-specific margin scales before aggregation.

## 4.7 Limitations and Future Work

Our error-aware data and S2 multimodal descriptions were generated with large multimodal language models, so their quality depends on the generator and may contain unverified noise. Although the generated error types are useful for controlled diagnostic evaluation, some negatives may mix multiple error patterns or be easier than naturally occurring human-written mistakes. Due to the limited workshop timeline, we could not manually verify all generated descriptions or fully rerun every model under exactly matched prompting settings, so the results should be interpreted as a practical challenge-system study rather than a fully curated benchmark.

Future work could add human verification for the generated error sets, annotate multi-label error types, and build a separate calibration split for more stable multi-judge fusion. A more complete study could also rerun all judges under the same S1/S2 protocol and analyze which error types benefit most from error-aware training.

# 5 Conclusion

We introduced EAPO, a framework for multimodal emotion preference learning that expands sparse human preference pairs with controlled error descriptions and combines independently adapted MLLM judges through calibrated continuous margins. The experiments indicate that agreement with human preference labels and robustness to plausible description errors are related but distinct capabilities: adaptation on the original pairs can improve preference prediction while reducing performance on controlled error subsets, whereas the complete error-augmented procedure better balances the two objectives. Calibrated Fusion further provides a practical means of using complementary judges without inter-model communication. These findings motivate evaluation protocols that assess both preference agreement and error robustness. Future work should validate the approach on independently annotated error sets and under fully matched inference and calibration protocols.

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